The emotional foundations of human prosociality

Amrisha Vaish* **and Yuhang Shu**
Department of Psychology, University of Virginia, Charlottesville, VA, United States
*Corresponding author. e-mail address: vaish@virginia.edu

Contents

1. Introduction 2
2. The emergence and nature of early prosociality 4
   2.1 Early prosocial behavior: Helping, comforting, and sharing 4
   2.2 Intrinsic prosocial motivation 6
   2.3 Summary 7
3. Initiating prosocial actions: The role of sympathy 7
4. Sustaining prosocial interactions 8
   4.1 Warm glow 9
   4.2 Gratitude 10
   4.3 Summary 12
5. Repairing prosocial relationships 13
   5.1 Guilt 13
   5.2 Forgiveness 14
   5.3 Summary 17
6. Conclusions and future directions 17
   6.1 A developmental proposal 17
   6.2 Caveats and open questions 18
   6.3 Conclusion 21
Funding 21
References 21

Abstract

Humans are fundamentally cooperative beings whose survival and success depend on prosocial behavior, including with non-kin and strangers. Even young children engage in prosocial behaviors such as helping, sharing, and comforting that are often intrinsically motivated. While evolutionary accounts explain why prosociality emerged, they leave open the proximate question of what motivates individuals, including young children, to engage in costly actions that benefit others. This chapter argues that evolution has equipped humans with affective mechanisms that motivate us to initiate, sustain, and repair prosocial interactions from early in development. Sympathy motivates us to initiate prosocial responses by creating

Advances in Child Development and Behavior
ISSN 0065-2407, https://doi.org/10.1016/bs.acdb.2026.07.004
Copyright © 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

concern for those in need. Warm glow (positive affect from acting prosocially) and gratitude sustain cycles of prosociality within and between individuals by making cooperation emotionally rewarding and motivating reciprocity. Guilt and forgiveness help repair damaged relationships, with guilt prompting transgressors to make amends and forgiveness enabling victims to restore trust and cooperative bonds. Drawing on behavioral, physiological, and cross-cultural research, we show that these mechanisms emerge within the first years of life and become increasingly flexible and context-sensitive with development. We conclude by outlining a developmental proposal for the emergence of prosocial emotions, discussing methodological and cultural challenges, and highlighting directions for future research. Together, the evidence suggests that early-emerging affective processes work in concert to enable even young children to navigate the cooperative landscape that defines human social life.

1. Introduction

Humans are tremendously social and cooperative beings. Our survival and success depend on our ability to live and work together—not just with kin and friends but also with non-kin and strangers. This interdependence emerged in our evolutionary history in response to environmental pressures that required humans to band together, collaborate, and cooperate to accomplish tasks they could not achieve alone (Johanson & Edey, 1990; Tomasello, 2018). Such cooperation provided long-term benefits, including increased chances of reciprocal help and the availability of future cooperative partners (Tomasello, 2018; Trivers, 1971). In turn, it meant we had to ensure the wellbeing of those with whom we were—or could be—interdependent.

This account helps explain the emergence of human prosociality at the *ultimate* level (i.e., why prosociality emerged evolutionarily). However, it does not explain what motivates individuals to behave prosocially at the *proximate* level (within a given interaction and within an individual's lifetime; Scott-Phillips et al., 2011). Prosocial behavior, which encompasses acts that benefit others such as helping, donating, sharing, and comforting, involves costs for the actor, who invests time, energy, or material resources to benefit another with no immediate return (Eisenberg et al., 2016). This raises a vital question: Why would an individual ever put aside their selfish interests to benefit others, particularly non-kin and strangers? What mechanisms might sustain such costly behavior?

One solution is that natural selection has shaped a range of psychological adaptations to enable us to override self-interest and prioritize our

interdependent and cooperative relationships (Bjorklund, 2018; Tooby & Cosmides, 1990; Vaish et al., 2016; Vaish & Hepach, 2020). In particular, we have evolved emotional mechanisms that drive and sustain our prosocial behavior.

This perspective aligns with functional theories of emotion, which conceptualize emotions as evolved adaptations that guide us to address challenges of adaptive and social significance (Campos et al., 1989; Keltner, Haidt, et al., 2006; Nesse, 1990). When individuals experience particular emotions, these affective states provide information about their circumstances and prompt responses suited to the situation at hand (Clore & Huntsinger, 2007; Keltner, Horberg, et al., 2006). Fear, for instance, arises in response to potential physical or social threats and produces physiological arousal and heightened vigilance that motivate individuals to protect themselves through fight, flight, or freeze responses (Nesse, 1990). Similarly, disgust is elicited by potential contagions and social violations and leads to feelings of revulsion, which motivate an individual to avoid or break off contact with the offending entity (Rottman, 2014; Rozin & Fallon, 1987; Tybur et al., 2013).

Building on this functional perspective, we propose that affective mechanisms similarly underpin and drive prosocial and cooperative behaviors at the proximate level (see, e.g., van Kleef & Lelieveld, 2022). Moreover, these affective mechanisms appear early in human development, enabling even the youngest members of our species to engage in prosocial actions (Bjorklund, 2018; Vaish & Hepach, 2020). Our overarching goal in this chapter is to examine what we consider to be central emotional mechanisms underlying early prosocial behavior.

Specifically, we are concerned with affective mechanisms that help initiate, sustain, and repair prosocial interactions. We propose that different mechanisms serve these complementary functions: The moral emotion of sympathy motivates us to initiate prosocial actions toward those in need; warm glow and gratitude help sustain prosociality; and guilt and forgiveness help us repair ruptures in our prosocial relationships. Recent developmental research demonstrates that these affective mechanisms begin to serve their prosocial functions from early in ontogeny, allowing even young children to reap the mutual benefits of cooperative living.

We begin by briefly reviewing evidence that prosocial behavior emerges early in ontogeny and that it is intrinsically rather than extrinsically motivated. We then discuss the affective mechanisms that move young children to initiate, sustain, and repair prosocial interactions, and conclude with a discussion of caveats, challenges, and future directions.

2. The emergence and nature of early prosociality

One of the most striking conclusions from recent developmental research is that even young children behave prosocially, and these capacities rapidly become more sophisticated with development (Svetlova et al., 2010; Warneken & Tomasello, 2013). Here we provide an overview of this work, focusing on helping, sharing, and comforting, and then present evidence that early prosocial behavior is underlain by intrinsic rather than extrinsic motivations.

2.1 Early prosocial behavior: Helping, comforting, and sharing

Instrumental helping—assisting another person in achieving an action-based goal—is the earliest prosocial behavior to emerge. In a classic study, Warneken and Tomasello (2006) found that 14–18-month-old infants in Germany helped adults complete their goals. With little prompting, children helped an experimenter by handing him out-of-reach objects or stacking books after the adult failed to do so himself. Even when children had to pay a cost by giving up an enjoyable activity, they still helped at high rates (Warneken et al., 2007). Critically, they did not perform these actions in control situations when the experimenter did not need help. These helping behaviors are evident around the same ages across cultures (Callaghan et al., 2011), suggesting that from early in life, children have a universal understanding of others' goals and the motivation to help them achieve those goals (Warneken & Tomasello, 2006). This early helping quickly becomes more sophisticated. By the preschool years, children engage in "paternalistic helping"—helping others achieve what is good for them rather than simply what they requested (Martin et al., 2016; Martin & Olson, 2013). Thus, young children understand not just what others want but what they need, and are motivated to help accordingly.

Also by the second year, children show comforting and caring behaviors toward those in distress. When 14–18-month-old infants witness their parent or another adult bump her knee and express pain and distress, many make some effort to intervene and alleviate the distress, through verbal comforting, direct aid (e.g., providing a bandage), and indirect helping (e.g., getting their mother to help) (e.g., Zahn-Waxler et al., 1992). By the third year, children intervene more readily and appropriately (Svetlova et al., 2010; Zahn-Waxler et al., 1992).

For young children, sharing with others poses greater challenges than helping or comforting, as it involves relinquishing resources valuable to

themselves. Nonetheless, sharing emerges early. In naturalistic observations, infants as young as 8 months may show or give toys to parents, siblings, and others, even when resources are low (e.g., Hay, 1979; Rheingold et al., 1976). Sharing becomes selective with development. For instance, 12-month-olds share more with peers and their own mothers than with the peers' mothers (Young & Lewis, 1979).

When children are endowed with resources in windfall scenarios and decide how to distribute them, early sharing is often limited: Preschool-aged children tend to keep most resources for themselves (Smith et al., 2013). Importantly, when children of the same ages obtain resources jointly through cooperation with others (reflecting the interdependence of early humans), they demonstrate a stronger sense of fairness (Hamann et al. 2011). With development, children become more generous and increasingly sensitive to social norms, sharing equally and rejecting unequal distributions, even at a cost to themselves (Benenson et al., 2007; Blake et al., 2015). For example, 6- to 8-year-olds preferred to throw away an additional resource—even one that could have gone to themselves—to avoid an unequal distribution (Shaw & Olson, 2012; see Zhang & Benozio, 2021, for cross-culture differences). Older children's sharing incorporates additional considerations beyond equality, such as merit and need (Baumard et al., 2012; Rizzo & Killen, 2016). By age 7, they compensate for disparities in initial resources by giving more to the disadvantaged recipient (Chai & He, 2017; Rizzo & Killen, 2016).

Over development, these forms of prosociality are all shaped by social norms about how people should behave. For instance, cultural norms about resource distribution shape children's understanding of fairness and their sharing (Blake et al., 2015; House, 2018). When sharing decisions are costly, children's responses diverge across societies around middle childhood, increasingly resembling the patterns observed in the adults within their own cultures (House, 2018; McAuliffe et al., 2017). Other social forces such as peer pressure also shape the degree and type of children's sharing (Chai et al., 2024). Similarly, children's comforting behaviors vary across contexts. For instance, 5-year-olds from Germany and Israel engaged in more overt comforting toward a distressed adult than children from Malaysia or Indonesia, despite showing comparable concern (Trommsdorff et al., 2007; see also Vreden et al., 2025). The researchers suggested that in cultures where hierarchical relationships are highly valued, children may hold back from assisting adults to avoid causing embarrassment. Early prosocial behaviors thus reflect not only children's emerging social capacities, but also culturally specific expectations about how caring for others should be expressed.

2.2 Intrinsic prosocial motivation

A critical question concerns what motivates children's early prosocial tendencies. One account suggests that children learn to behave prosocially because they anticipate external rewards (Bar-Tal, 1982). A contrasting view suggests that evolutionary pressures have made us genuinely invested in the wellbeing of those with whom we are interdependent, such that prosocial actions are intrinsically motivated (Aknin et al., 2018; Tomasello, 2018). One method for distinguishing between these possibilities examines how material rewards affect prosocial behavior. The "over-justification effect" suggests that extrinsic incentives can diminish the intrinsic motivation to perform a behavior (Deci et al., 1999). Applying this logic to children's helping, Warneken and Tomasello (2008) found that 20-month-olds who received material rewards for helping were subsequently less likely to help than those who received social rewards or no rewards. Similarly, 3-year-olds who were materially rewarded for costly sharing later showed less sharing than children who received social rewards or no rewards (Ulber et al., 2016). Material rewards thus undermine young children's intrinsic prosocial motivation.

Interestingly, social rewards such as praise and encouragement may *support* prosocial behaviors when they first emerge (13–14 months), though they have no impact just a few months later (15–17 months; Dahl et al., 2017). This does not mean, however, that infants initially act prosocially *in order to* obtain social rewards. Moreover, this work and the over-justification research show converging evidence that by at least 1.5 years, children's prosocial motivation is largely intrinsic.

Other research shows that children are motivated to see those in need receive help, regardless of who provides the help. Studies using pupil dilation—a physiological marker of internal arousal (Bradley et al., 2008; Hepach & Westermann, 2016)—find that children show increased pupil dilation upon seeing someone in need of help, with larger increases predicting faster helping (Hepach et al., 2019). In one study using this method (Hepach et al., 2012), 2-year-olds first saw a person in need of help, which elicited pupil dilation (suggesting an increase in prosocial motivation). Children could then help the person themselves, were prevented from helping but observed someone else provide help, or observed the person not being helped. The study revealed that children's pupil dilation decreased similarly when they provided help themselves or observed another person provide help, but remained elevated when no

help was provided. Children were thus primarily motivated to see the person in need receive help, not gain recognition or rewards for providing help themselves. Further research has ruled out other alternatives, such as that children simply like completing action sequences or restoring physical order (Hepach et al., 2016), bolstering the conclusion that early prosocial actions are driven by a genuine motivation to fulfill others' needs.

2.3 Summary

Even young children engage in prosocial behaviors including helping, comforting, and sharing. These start out relatively simple but with development, they increase in complexity and flexibility. Early prosociality is grounded in genuine regard for others' welfare, providing a foundational prosocial motivator. Of course, children (and adults) are not always prosocial when given the opportunity, nor are all children (or adults) equally prosocial (e.g., Newton et al., 2016). The key point, however, is that from early in life, children are clearly and intrinsically motivated to engage prosocially, often at some cost. The question we turn to next is: What affective mechanisms underlie these early-emerging prosocial behaviors?

3. Initiating prosocial actions: The role of sympathy

Because prosocial actions often require individuals to incur costs without receiving immediate benefits, a central question is what drives people—including young children—to initiate such behavior. We propose that a key affective mechanism is empathic responding. This broad category includes *empathy*, defined as sharing or matching another person's emotional state, and *sympathy*, or feelings of concern for a person in need or distress (Batson, 1991; Eisenberg et al., 2006). Although closely related, empathy and sympathy are distinct, and it is sympathy rather than empathy that promotes prosocial action (Jordan et al., 2016). Sympathy is thus argued to be a core feature of human nature: a species-wide phenomenon that gives observers quick and direct access to others' suffering and motivates prosocial behavior to alleviate that suffering (Hoffman, 2000, 2007).

Research in diverse cultural contexts indicates that sympathy functions as an early prosocial motivator. By the second year, toddlers respond to others' distress with concerned facial expressions and verbalizations, and individual differences in their concern predict their levels of prosocial behavior

(Eisenberg & Fabes, 1990; Huang et al., 2017; Trommsdorff et al., 2007; Zahn-Waxler et al., 1992). Indeed, concern for others' suffering is evident within the first few months after birth; although infants at this age rarely act prosocially, their early concern predicts their prosocial behavior in the second year (Davidov et al., 2025; Grossmann et al., 2018). The basic capacity to care about others is thus present early and motivates prosocial actions once children acquire the necessary cognitive, social, and motor abilities.

Initially, concern is elicited by clear, observable signals of distress, such as crying or pain. In the second and third years, cognitive advances lead to more nuanced sympathy that no longer depends on explicit cues (Vaish, 2016). A key skill underlying this shift is affective perspective taking: the ability to take another's perspective and infer their emotional state. As this capacity emerges around 18 months, children can sympathize even without observable distress, such as when a victim is stoic or absent, and this concern motivates their prosocial behavior toward the victim (Vaish et al., 2009; see also Chiarella & Poulin-Dubois, 2015).

Around the same ages, sympathy becomes context sensitive. In one study, 3-year-olds showed greater sympathy and helping toward an adult displaying justified than unjustified distress, and the more sympathy children expressed, the faster they helped the adult (Hepach et al., 2012). Even 18-month-olds showed greater concern for a justifiably than unjustifiably distressed adult (Chiarella & Poulin-Dubois, 2013). Thus, during the second year, sympathy becomes increasingly flexible and robustly motivates prosocial actions. Such flexibility makes evolutionary sense, as a reliable prosocial motivator should be one that has a deep biological basis but is also flexible based on cues and context (Hoffman, 2000). Sympathy fits this bill from early in life (see Vaish, 2016).

In sum, developmental work reveals an early-emerging, flexible concern for others' suffering that motivates prosocial behavior. As children's cognitive capacities develop, their sympathy becomes increasingly flexible and context sensitive, allowing them to respond appropriately to others' needs. Sympathy thus provides a foundational and reliable affective basis for initiating prosocial behavior toward those in need.

4. Sustaining prosocial interactions

Beyond initiating prosocial actions, additional mechanisms are required to sustain prosociality across time and repeated interactions. We focus here on

mechanisms that help sustain *cycles of prosociality* (Beeler-Duden & Vaish, 2024), that is, that one prosocial act is more likely to lead to a second prosocial act, then a third, and so forth. These cycles can occur intra-individually (a person who acts prosocially is more likely to act prosocially again in the future) and inter-individually (recipients of prosocial acts are more likely to act prosocially in turn). We propose that the intra-individual cycle is supported by warm glow, and the inter-individual cycle is promoted by gratitude.

4.1 Warm glow

To motivate us to put aside our self-interests to benefit others, humans are thought to have evolved the tendency to experience prosocial actions as inherently rewarding, that is, to experience a "warm glow" after behaving prosocially (Aknin et al., 2012, 2018). The "warm-glow giving" framework proposes that warm glow in turn encourages people to engage in additional prosocial actions, creating a positive feedback loop that sustains the intra-individual cycle of prosociality (Aknin et al., 2018; Andreoni, 1989, 1990).

Research with adults supports this account. Allocating one's money to benefit others increases happiness across cultures (Aknin et al., 2012, 2013), and sharing and charitable giving engage neural systems linked to reward processing (Harbaugh et al., 2007; Zaki et al., 2011). Additionally, adults who engage more frequently in prosocial acts report higher overall well-being (Dunn et al., 2008). Thus, the emotional payoff of acting prosocially may serve as a driver of sustained prosocial engagement among adults.

Similar rewarding effects are seen in early childhood. In one study, 2-year-olds showed more positive facial expressions when giving treats to others than receiving treats, and strikingly, children were happier after costly than non-costly sharing (Aknin et al., 2012). Toddlers and preschool-aged children also showed greater positive affect after instrumental helping and after helping alleviate another's distress (Song et al., 2020). This warm glow effect is evident among young children across diverse cultural contexts (e.g., Aknin et al., 2012, 2015; Paulus & Moore, 2017; Song et al., 2020).

Recent methodological innovations have shed further light. One approach involves measuring changes in children's upper-body posture (using motion-sensing technology such as Microsoft© Kinect) as an index of emotional valence, with more upright upper-body posture indicating more positive affect (Gerdemann et al., 2025). Hepach et al. (2017b) found that 2-year-olds showed comparable increases in postural elevation when they

accomplished a goal for themselves and for someone else. In contrast, postural elevation decreased when children's actions failed to benefit anyone. Sivertsen et al. (2024) documented greater postural elevation after preschoolers engaged in costly than non-costly helping, bolstering prior behavioral findings. This burgeoning body of work provides compelling evidence that acting prosocially elicits positive affect in early childhood.

Comparatively less research has directly examined a core prediction of warm glow: that it subsequently motivates future prosocial behavior. Some work with adults has provided initial evidence for this positive feedback loop (e.g., Aknin et al., 2012; Snippe et al., 2018), and emerging work with children is suggestive as well. Paulus and Moore (2017) asked preschool-aged children to anticipate how they would feel after sharing resources. The positive emotions that children anticipated in these hypothetical scenarios predicted their later sharing in a resource allocation task. Though interesting, further research is needed to experimentally test the self-reinforcing function of warm glow in development.

More research is also needed to understand the contexts in which children experience warm glow. Several factors shape adults' warm glow such as similarity, familiarity, and strength of the social relationship (Aknin et al., 2011; Bohnet & Frey, 1999; Small & Loewenstein, 2003). Whether and how these factors impact young children's experience of warm glow is an important direction for future research.

In sum, acting prosocially elicits positive affect from early in life and across cultural contexts. This points to warm glow as a potentially universal affective mechanism that intrinsically rewards prosocial actions and helps sustain intra-individual cycles of prosociality.

4.2 Gratitude

Vital to successful cooperation is the ability to recognize good partners who have our interest at heart, and to maintain these favorable partnerships. Gratitude is thought to serve just this function. Gratitude is a positive emotion that arises from the perception that one has benefited from the goodwill of another (McCullough et al., 2001). It is theorized to help individuals recognize and reinforce benevolent actions, support reciprocal exchange, and strengthen social bonds (Algoe, 2012; Nowak & Sigmund, 2005).

Central to these accounts is the idea that gratitude motivates reciprocity, effectively sustaining cycles of cooperation that underlie human social life (Bonnie & de Waal, 2004; Trivers, 1971). Research with adults robustly supports this function of gratitude: Individuals who are made to

feel grateful towards a benefactor are subsequently more likely to help the benefactor compared with those who did not experience gratitude (Ma et al., 2017). Importantly, gratitude predicts greater prosocial responding than either general positive mood or positive evaluations of the benefactor, highlighting its unique motivational role (Tsang, 2006b, 2006a). Thus, gratitude motivates us to reciprocate toward those who showed us goodwill.

Until recently, relatively little attention had been paid to understanding the early development of gratitude and its reciprocal function. Some research using naturalistic observations and hypothetical scenarios suggested that gratitude emerges around 6–7 years of age and develops slowly thereafter (Gleason & Weintraub, 1976; Graham, 1988; Poelker & Kuebli, 2014). However, as much of this work relied on children's verbal capacities or use of verbal routines such as "thank you," it likely underestimated younger children's capacities for gratitude.

More recent research within a functionalist framework has used interactive situations and behavioral measures, and revealed remarkably early gratitude-like patterns. For example, 3-year-olds (but not 2-year-olds) shared more resources with a partner who previously shared with them than one who did not (Warneken & Tomasello, 2013), and even 2-year-olds chose to help their benefactor over a non-benefactor (Dunfield & Kuhlmeier, 2010). However, reciprocity alone does not demonstrate gratitude, as children may act prosocially simply due to positive affect about receiving benefits or positive evaluations of generous individuals in general. To establish gratitude as a distinct motivator of early prosociality, it is necessary to show that children's reciprocity is in response to the two core components of gratitude: personal receipt of a benefit and the benefactor's good intentions (McCullough et al., 2008; Tsang, 2006a).

Newer research has tackled this question. In one study, 3-year-old German children either received resources from a helpful puppet who intentionally shared with them or from an indifferent puppet who did not convey the intention to help, or they observed another individual receive resources from the helpful puppet (Vaish et al., 2018). Children later shared most generously with the puppet who had intentionally helped them. Thus, children's reciprocity was driven not by general positive mood or positive evaluations of generous individuals but by the key conditions of gratitude. Similar findings emerged among 3–6-year-old Israeli children (Shoshani et al., 2020, 2021). Thus, by age 3, a nascent gratitude seems to motivate selective reciprocity toward reliable cooperative partners.

Even younger children may have a budding sense of gratitude. Although toddlers do not show overt reciprocal helping, receiving help appears to alter their underlying prosocial motivation. Hepach and colleagues (2019) used pupil dilation to measure 1.5- to 2-year-olds' prosocial motivation. They found that if someone in need had previously helped the children, then children's arousal remained elevated if they could not help the benefactor themselves, indicating a specific desire to actively return help. Thus, before age 3, children may already be sensitive to another's goodwill toward them and motivated to uphold that relationship.

Beyond motivating direct reciprocity, gratitude also supports a surprising form of prosociality: paying it forward (upstream reciprocity). This helps initiate new inter-individual cycles of prosociality, which is critical for expanding cooperative networks, particularly to non-kin (Nowak & Roch, 2006). Among adults, upstream reciprocity is also predicted by gratitude rather than general positive affect (Bartlett & DeSteno, 2006; Tsang, 2006a, 2007), and recent work suggests similar effects in early childhood. Beeler-Duden and Vaish (2020) found that after 4-year-olds received help on a challenging task, they were more likely to share resources with a new child (not the helper) than if they had not received help, and children's generosity was predicted by their gratitude-related responses rather than reported happiness. Shoshani et al. (2020) similarly found that inducing gratitude in 4- to 6-year-old Israeli children increased their helping toward strangers.

Thus, an early sense of gratitude appears and serves its key prosocial functions within the first few years. By around 2–3 years, gratitude motivates children to reciprocate benefits toward their benefactors, thereby maintaining inter-individual prosocial cycles. By around 4 years, gratitude may also motivate children to pay benefits forward, enabling new prosocial associations. In these ways, early gratitude plays an important role in supporting the prosocial cycles that are essential for large-scale and sustained human cooperation.

4.3 Summary

Sustaining prosocial behavior requires affective mechanisms that reinforce repeated prosociality within and between individuals. The literature supports two such mechanisms: warm glow supports intra-individual cycles of prosociality, and gratitude supports inter-individual cycles. Growing evidence from children across cultures indicates that both mechanisms are active and jointly help sustain and expand cycles of prosociality from early in life, thus supporting large-scale and enduring human cooperation.

5. Repairing prosocial relationships

Despite the prosocial motivations that move humans to take care of one another, our cooperative relationships are inevitably damaged from time to time by transgressions, both big and small. In such instances, it is vital that we repair the ruptured relationships and continue reaping the mutual benefits the relationships provide. We propose that this repair is facilitated by two complementary affective mechanisms: Guilt motivates transgressors to make amends and seek reconciliation, while forgiveness allows victims to reestablish trust and restore the cooperative bond.

5.1 Guilt

When individuals cause harm to others, they typically experience guilt: an aversive social emotion that arises from the recognition that one's actions have harmed others (Nelissen & Zeelenberg, 2009). This emotional response differs from merely witnessing a person who was harmed, which elicits sympathy and prosocial behavior. Guilt involves the intersection of sympathy for a harmed individual and the awareness of having *caused* that harm (Hoffman, 1982). Neither element alone constitutes guilt, yet their combination produces a powerful motivational force directed toward relationship repair.

Guilty feelings prompt transgressors to focus on the damage they have caused and motivate compensatory actions toward their victims (Keltner, 1995). The prosocial functions of guilt are well-established in adults across cultures (Ketelaar & Tung Au, 2003; Molho et al., 2025; Tang et al., 2019). For instance, participants who experienced guilt after acting selfishly subsequently increased their cooperative behavior, whereas those who did not feel guilty did not show this shift (Ketelaar & Tung Au, 2003). Guilt thus operates as a relationship maintenance mechanism, helping us identify when we have damaged social bonds and spurring us to restore those connections (Baumeister et al., 1994).

These reparative functions emerge surprisingly early (Vaish, 2018). In observational studies, toddlers as young as 2 years display guilt-like responses, such as taking responsibility and attempting repairs, after minor transgressions (Drummond et al., 2017; Kochanska et al., 1995). However, these studies leave unclear whether children experienced guilt or were instead responding to isolated components: sympathy for the victim, or the desire to fix what they broke without concern for the victim. To disentangle these possibilities, Vaish et al. (2016) systematically varied whether children caused harm,

someone else caused identical harm, and whether the outcome was harmful at all. Three-year-olds showed significantly more reparative behavior—both physically and verbally—in the guilt-inducing condition (when they had caused harm) compared to the other scenarios. Two-year-olds showed sympathy-driven responses to others' distress but did not display this guilt-specific pattern.

Complementary research using physiological measures provides additional insight. When children caused accidental harm, 3-year-olds' (and more weakly, even 2-year-olds') pupil dilation decreased only when they were personally able to repair the damage; it remained elevated if another person stepped in (Hepach et al., 2017a). Thus, causing harm fundamentally transforms children's prosocial motivation. Rather than simply wanting to see the victim helped (as when they are simply bystanders: Hepach et al., 2012), children who have caused harm specifically want to be the ones who help, reflecting an emerging understanding that they must personally address the relationship breach.

Thus, by age 3, and perhaps in nascent form even earlier, children show guilt-like responses: They recognize when they have caused harm and are motivated to repair it. Notably, young children rarely apologized conventionally by saying "sorry." Rather, their guilt-consistent responses were evident in functional ways, such as physical and verbal reparative attempts, accepting responsibility, or changes in underlying prosocial motivation. Thus, before learning the social rituals of verbal apologies, young children demonstrate functional guilt responses that motivate and guide their reparative social behavior. This helps them navigate inevitable social transgressions, providing a mechanism for restoring valuable relationships.

As children mature, their responses extend beyond this dyadic form of guilt to collective responsibility. By age 5, children in Germany and China are more willing to accept responsibility and make reparative efforts for harm caused by their in-group than out-group members (Over et al., 2016; Zhang & Li, 2026). This developmental trajectory positions guilt as a foundational element in the maintenance of many kinds of cooperative relationships throughout the lifespan.

5.2 Forgiveness

While guilt motivates transgressors to make amends, repairing damaged relationships requires a complementary response from victims: forgiveness. Forgiveness is the restoration of positive feelings toward one's transgressors and it motivates reconciliation (McCullough et al., 1997). It thus represents

an adaptive solution to a fundamental social problem. While avoiding or seeking revenge against transgressors could prevent future exploitation, these responses can provoke retaliation and permanently sever potentially valuable partnerships. Forgiveness circumvents these risks by reducing aggression and restoring prosocial motivations (McCullough et al., 2013).

Critically, if forgiveness evolved to optimally protect valuable relationships, it should be sensitive to cues indicating whether forgiving a particular transgressor is likely to yield net benefits or invite further harm (Burnette et al., 2012; McCullough et al., 2014). Research with adults confirms that forgiveness has boundaries that track the risk of future exploitation (McCullough et al., 2013). Two primary cues shape these boundaries: transgressors' remorse, which signals awareness of wrongdoing and future trustworthiness (Fehr et al., 2010; Keltner et al., 1997), and transgressors' intentions, with intentional transgressions indicating dispositional traits that predict future harm (Davis & Gold, 2011; Struthers et al., 2008).

The development of forgiveness has traditionally been studied using a cognitive-developmental model, focused on how reasoning about forgiveness changes during adolescence, driven by cognitive growth (Enright et al., 1992). This focus has precluded younger children, whose verbal and reasoning skills are limited. Yet if forgiveness facilitates social relationships, it may emerge in early childhood as children begin to form cooperative ties. Recent research has adopted this functionalist perspective, using child-friendly behavioral tasks and revealing that the capacity to forgive—and do so selectively—emerges in the preschool years.

Most of this work has examined children's responses to remorse. When 4- and 5-year-olds witnessed transgressions against others, they blamed apologetic transgressors less, liked them more, and preferred to cooperate with them than with non-apologetic transgressors (Smith et al., 2010; Vaish et al., 2011). By age 5 (but not 4), children showed the same pattern toward transgressors who showed remorse even without standard apology scripts such as "sorry," suggesting that by 5, children understand the cooperative meaning of remorse (Vaish et al., 2011). Very similar findings emerged when 4- and 5-year-olds were themselves the victims (Oostenbroek & Vaish, 2019b), and 6-year-olds who had been harmed reported feeling better after receiving apologies (Drell & Jaswal, 2016; Smith & Harris, 2012). By these ages, children also grasp that forgiveness helps restore relationships, and prefer those who forgive over those who do not (Oostenbroek & Vaish, 2019a; Toda et al., 2024).

The perceived intentions underlying a transgression also shape forgiveness, as intentional harms signal ill will and thus heightened risk of future exploitation. Adults are less forgiving of intentional than accidental transgressions (Fehr et al., 2010). Similarly, young school-age children assign more blame and show less forgiveness following intentional than accidental transgressions (Amir et al., 2021; Darby & Schlenker, 1989; McElroy et al., 2023). Thus, quite young children have the capacity to forgive and, like adults, their forgiveness is sensitive to transgressors' remorse and intentions.

Children's forgiveness also integrates remorse and intentions with broader social information. Vaish and Oostenbroek (2022) found that 5-year-olds selectively forgave transgressors who signaled greater cooperative value: They were more forgiving of remorseful in-group than out-group transgressors (in line with in-group preference), but this pattern reversed when only the out-group transgressor was remorseful (see also Mulvey et al., 2022). Similarly, 4–6-year-olds were more forgiving of accidental transgressors and authority figures, but when these cues conflicted, children prioritized intentions over authority (Cheng et al., 2025). Early forgiveness is thus calibrated to balance relationship benefits against exploitation risk.

Finally, forgiveness should exhibit not only boundaries but also the flexibility to stretch those boundaries when appropriate (Burnette et al., 2012; McLaughlin et al., 2024, 2025). For instance, severe or intentional transgressors are expected to go beyond perfunctory apologies and provide elaborate remorse—expressing empathy, offering compensation, and signaling renewed commitment—which adults perceive as more sincerely remorseful and justice-restoring (Fehr & Gelfand, 2010; Ohtsubo & Watanabe, 2009; Scher & Darley, 1997). When elaborate apologies are provided, even egregious transgressors may be brought back into the cooperative fold.

New work shows that children are similarly sensitive to quality of remorse. Five- and 6-year-olds preferred transgressors who apologized and provided a reason over those who only apologized, and preferred transgressors who provided distinct reasons for repeated harms (Waddington et al., 2022, 2023). They also evaluated and shared more resources with an intentional transgressor who offered an elaborate apology that expressed regret and empathy versus a simple apology (Clayton & Vaish, 2026). By attending to the depth and quality of remorse, children can stretch the usual limits of forgiveness to restore cooperation even after more egregious transgressions.

5.3 Summary

When our prosocial relationships are damaged by transgressions, we must repair them to continue to reap their mutual benefits. Guilt motivates transgressors to make amends, and recent work shows that it serves this function as early as age 3. Complementing guilt, forgiveness allows victims to restore trust and cooperative bonds. By late preschool age, children selectively forgive based on cues of transgressors' trustworthiness, particularly remorse and intentionality. Children are also sensitive to the depth and quality of remorse. Together, these mechanisms enable young children to navigate social ruptures and preserve valuable partnerships while protecting against exploitation.

6. Conclusions and future directions

In this final section, we first outline a developmental proposal synthesizing the emergence of prosocial emotions. We then conclude with some caveats, challenges, and open questions for this area of inquiry.

6.1 A developmental proposal

Prosocial motivation emerges early in childhood and becomes increasingly sophisticated. Within the first 18 months, infants exhibit prosocial behaviors driven by sympathy. By ages 2–3, children show elevated physiological arousal when unable to help their benefactors or repair harm that they caused, though these motivations do not consistently translate into overt helping behaviors. Around 3–4 years, a crucial shift occurs as children direct more prosocial behavior toward those they harmed and those who showed them goodwill. This progression likely reflects social-cognitive advances such as self-awareness, theory of mind, and executive functioning, which allow children to act effectively on their motivations. Further, gratitude and guilt are more complex than sympathy because they require advanced social-cognitive capacities, including adhering to social standards and evaluating oneself and others accordingly (Schmidt et al., 2019; Tomasello, 2016). This "normative turn" occurs between 2 and 3 years, corresponding to the first evidence of these complex emotions. By ages 3–4, children have growing awareness that others can observe and evaluate their actions (Evans & Lee, 2013; Kelsey et al., 2018). Children of these ages may recognize the importance of demonstrating commitment through reparative and reciprocal actions,

transforming internal motivations into observable behaviors. With the emergence of these complex affective mechanisms, children can now actively maintain their prosocial relationships.

By 4–5 years, children respond appropriately to others' displays of guilt and remorse (as well as displays of gratitude and forgiveness; Oostenbroek & Vaish, 2019a; Toda et al., 2024; Vaish & Savell, 2022). This later emergence may reflect the fact that social emotions such as guilt are conveyed through complex, context-dependent facial, verbal, and behavioral actions, and learning to recognize and understand them may take time and social experience (Pons et al., 2003). Together, these rapidly growing abilities and their interactions allow children in their late preschool to early school years to identify reliable and trustworthy cooperative partners.

Notably, the emergence of complex prosocial emotions coincides with children's increasing interactions with peers. Unlike child-caregiver bonds, which are asymmetrical and provide unconditional care, peer relationships are symmetrical and reciprocal: Children risk social consequences such as peer avoidance or exclusion if they fail to show gratitude and reciprocity or guilt and repair. We thus propose that it is relationships with peers rather than caregivers that create the selection pressures necessary for prosocial emotions to fully emerge (see Hepach & Vaish, 2020). This proposal is tentative, however, and requires systematic empirical investigation.

6.2 Caveats and open questions

6.2.1 Methodological advances and challenges

Studying complex prosocial emotions poses significant methodological challenges because these emotions lack distinct facial expressions and are expressed multimodally. This challenge is compounded among young children, who have limited verbal and introspective abilities. Consequently, many studies focus on observable behaviors and whether those behaviors align with the predicted functions of the emotions. Of course, early forms of the emotions may not serve all later functions. For example, whereas adult guilt extends to collective guilt for the misdeeds of close others (Doosje et al., 1998), early guilt is limited to dyadic harm as children lack the group-mindedness needed for collective guilt until 4–5 years (Over et al., 2016). We propose that prosocial emotions first serve their most essential functions, such as reciprocity for gratitude and repair for guilt. These functions become more complex with development, supported by advances in cognition and social understanding. Importantly, the absence of more advanced functions

in early childhood does not mean the absence of the emotions. Future research should integrate implicit, developmentally continuous measures of prosocial motivation such as pupil dilation, postural elevation, and cardiovascular responding (Gerdemann et al., 2025; Hepach et al., 2019; Miller, 2018), combined with behavioral, facial, and self-report assessments in older participants to obtain converging evidence that allows for more confident conclusions.

6.2.2 Other prosocial emotions

Beyond the emotions we focused on, many other affective mechanisms support our complex social lives. Embarrassment, for instance, occurs after social convention violations and motivates individuals to acknowledge mistakes and attend more to the social norm in the future. Showing embarrassment elicits affiliative responses such as sympathy and helping from observers, thereby restoring smooth social interaction (Keltner et al., 1997). Yet little is known about the ontogeny of these functions of embarrassment. Research on shame is similarly limited. Unlike guilt, which focuses on one's harmful *actions*, shame is the feeling of a flawed *self*, and is typically linked to withdrawal and reduced prosociality (Tangney & Dearing, 2002). However, shame may, under some conditions, motivate reparative attempts (Leach & Cidam, 2015). Children show shame-like withdrawal after transgressing by age 2 (Drummond et al., 2017), but it remains unclear whether shame could also motivate children to repair. Pride, a positive emotion that motivates social proximity, may also reinforce prosociality: By 3–4 years, children show more pride following spontaneous than prompted helping, which, like warm glow, may reinforce future prosocial behavior (Ross, 2017; Tracy et al., 2005). Further research is needed to explore whether these (and other) affective mechanisms also assist with initiating, maintaining, and repairing prosocial interactions.

6.2.3 Self-serving and multiple motivations

Although humans are predisposed to care about others' welfare, prosocial behavior can also be driven by self-serving motives such as reputation management, which become increasingly prominent over development (Grueneisen & Warneken, 2022). By age 5, children behave more generously when observed, and show more positive affect when they provide help themselves than observe someone else providing help, consistent with emerging reputational concerns (Engelmann & Rapp, 2018; Hepach et al., 2023).

Crucially, self-interested motivations do not preclude genuinely other-oriented motivations. Multiple motivations can co-exist and jointly support prosociality, reflecting evolutionary selection for diverse pathways toward adaptive social behaviors (Vaish & Tomasello, 2022).

We must also consider non-affective mechanisms. People may, for instance, act prosocially based on moral principles or deliberative judgments, independent of affect (e.g., Eisenberg et al., 2016). These mechanisms are as important to study as affective processes, and may be more amenable to interventions to promote prosocial behavior (Bloom, 2017).

6.2.4 Cultural influences

The developmental research we described—particularly on complex mechanisms such as gratitude and guilt—was predominantly conducted in WEIRD populations (Henrich et al., 2010). Yet cultural contexts significantly influence prosocial motivators and their outcomes. This has been demonstrated compellingly for early sympathy. Trommsdorff et al. (2007) found that sympathy predicted young children's prosocial behavior toward an adult in Germany, Israel, Indonesia, and Malaysia, but the association was weaker in the Southeast Asian contexts, perhaps because children in those contexts refrained from acting on their sympathy in order not to cause the adult to 'lose face.' Kärtner et al. (2010) further highlighted that the developmental pathway from sympathy to prosocial behavior is shaped by sociocultural context: Sympathy appeared to motivate children's prosociality through parents' relational socialization goals (e.g., interpersonal responsiveness) in India but through autonomous goals in Germany.

Cultural variations likely exist for other prosocial motivators. For example, culture influences conceptions of "ideal affect" (Tsai et al., 2006), which may impact how rewarding people find warm glow. Further, benefactor motives, considered critical for gratitude in WEIRD contexts, may be less significant in non-WEIRD contexts of tight knit, enduring social units (Barrett et al., 2016; see Henrich, 2020). Such variations likely impact the development of gratitude. Indeeed, Kärtner et al. (2016) found that German and Indian mothers scaffolded toddlers' responses to a gift differently, with German mothers maximizing positive affect and focusing on the gift and Indian mothers emphasizing social dimensions and acknowledging the gift giver. To fully understand these and other prosocial motivators, future research must be firmly grounded in cultural contexts.

6.3 Conclusion

The developmental evidence indicates that human prosocial behavior is rooted in early-emerging, other-oriented affective processes. These prosocial emotions appear remarkably early in life: Sympathy is observable within the first year, nascent forms of warm glow, gratitude, and guilt by ages 2–3, and more complex manifestations by ages 4–5. Multiple affective mechanisms work in concert to initiate, sustain, and repair cooperative relationships. This affective architecture allows even young children to navigate the cooperative landscape and participate in the social networks that define human life.

From an evolutionary perspective, human interdependence created selection pressures favoring cooperation, and early-emerging affective mechanisms serve as the proximate psychological processes through which such cooperation is realized in development. Understanding when and how prosocial emotions emerge illuminates both the psychological foundations of human cooperation and the mechanisms through which evolution has equipped our species for cooperative group life. The toddler who comforts a distressed peer, feels joy after sharing, or attempts to repair a relationship after causing harm is already a participant in the cooperative networks that make human societies possible.

Funding

The preparation of this chapter was supported in part by a grant from the John Templeton Foundation (Grant No. 62641) to Amrisha Vaish.

References

Aknin, L. B., Barrington-Leigh, C. P., Dunn, E. W., Helliwell, J. F., Burns, J., Biswas-Diener, R., ... Norton, M. I. (2013). Prosocial spending and well-being: Cross-cultural evidence for a psychological universal. *Journal of Personality and Social Psychology*.

Aknin, L. B., Broesch, T., Hamlin, J. K., & Van De Vondervoort, J. W. (2015). Prosocial behavior leads to happiness in a small-scale rural society. *Journal of Experimental Psychology: General, 144*(4), 788–795. https://doi.org/10.1037/xge0000082.

Aknin, L. B., Hamlin, J. K., & Dunn, E. W. (2012). Giving leads to happiness in young children. *PLoS One,* 7(6), e39211. https://doi.org/10.1371/journal.pone.0039211.

Aknin, L. B., Sandstrom, G. M., Dunn, E. W., & Norton, M. I. (2011). It's the recipient that counts: Spending money on strong social ties leads to greater happiness than spending on weak social ties. *PLoS One, 6*(2), https://doi.org/10.1371/journal.pone.0017018.

Aknin, L. B., Van de Vondervoort, J. W., & Hamlin, J. K. (2018). Positive feelings reward and promote prosocial behavior. *Current Opinion in Psychology, 20*, 55–59. https://doi.org/10.1016/j.copsyc.2017.08.017.

Algoe, S. B. (2012). Find, remind, and bind: The functions of gratitude in everyday relationships. *Social and Personality Psychology Compass, 6*(6), 455–469. https://doi.org/10.1111/j.1751-9004.2012.00439.x.

Amir, D., Ahl, R. E., Parsons, W. S., & McAuliffe, K. (2021). Children are more forgiving of accidental harms across development. *Journal of Experimental Child Psychology, 205*. https://doi.org/10.1016/j.jecp.2020.105081.

Andreoni, J. (1989). Giving with impure altruism: Applications to charity and ricardian equivalence. *Journal of Political Economy, 97*(6), 1447–1458.

Andreoni, J. (1990). Impure altruism and donations to public goods: A theory of warm-glow giving. *Economic Journal, 100*, 464–477. https://doi.org/10.2307/2234133.

Barrett, H. C., Bolyanatz, A., Crittenden, A. N., Fessler, D. M. T., Fitzpatrick, S., Gurven, M., ... Laurence, S. (2016). Small-scale societies exhibit fundamental variation in the role of intentions in moral judgment. *Proceedings of the National Academy of Sciences, 113*(17), 4688–4693. https://doi.org/10.1073/pnas.1522070113.

Bar-Tal, D. (1982). Sequential development of helping behavior: A cognitive-learning approach. *Developmental Review, 2*(2), 101–124. https://doi.org/10.1016/0273-2297(82)90006-5.

Bartlett, M. Y., & DeSteno, D. (2006). Gratitude and prosocial behavior: Helping when it costs you. *Psychological Science, 17*(4), 319–325. https://doi.org/10.1111/j.1467-9280.2006.01705.x.

Batson, C. D. (1991). *The altruism question: Toward a social-psychological answer*. Lawrence Erlbaum Associates, Inc, ix, 257.

Baumard, N., Mascaro, O., & Chevallier, C. (2012). Preschoolers are able to take merit into account when distributing goods. *Developmental Psychology, 48*(2), 492–498. https://doi.org/10.1037/a0026598.

Baumeister, R. F., Stillwell, A. M., & Heatherton, T. F. (1994). Guilt: An interpersonal approach. *Psychological Bulletin, 115*(2), 243–267. https://doi.org/10.1037/0033-2909.115.2.243.

Beeler-Duden, S., & Vaish, A. (2020). Paying it forward: The development and underlying mechanisms of upstream reciprocity. *Journal of Experimental Child Psychology, 192*. https://doi.org/10.1016/j.jecp.2019.104785.

Beeler-Duden, S., & Vaish, A. (2024). Feeling good and feeling thankful: The role of positive emotions in sustaining early prosocial behavior. *The Routledge International Handbook of Multidisciplinary Perspectives on Character Development: Conceptualizing and Defining Character, 1*, Routledge. 587–610. https://doi.org/10.4324/9781003251248-32.

Benenson, J. F., Pascoe, J., & Radmore, N. (2007). Children's altruistic behavior in the dictator game. *Evolution and Human Behavior, 28*(3), 168–175. https://doi.org/10.1016/j.evolhumbehav.2006.10.003.

Bjorklund, D. F. (2018). How children invented humanity. *Child Development, 89*(5), 1462–1466. https://doi.org/10.1111/cdev.13020.

Blake, P. R., McAuliffe, K., Corbit, J., Callaghan, T. C., Barry, O., Bowie, A., ... Warneken, F. (2015). The ontogeny of fairness in seven societies. *Nature, 528*(7581), 258–261. https://doi.org/10.1038/nature15703.

Bloom, P. (2017). *Against empathy: The case for rational compassion*. Random House.

Bohnet, I., & Frey, B. S. (1999). Social distance and other-regarding behavior in dictator games: Comment. *American Economic Review, 89*(1), 335–339. https://doi.org/10.1257/aer.89.1.335.

Bonnie, K. E., & de Waal, F. B. M. (2004). Primate social reciprocity and the origin of gratitude. *The psychology of gratitude*. Oxford University Press. 213–229. https://doi.org/10.1093/acprof:oso/9780195150100.003.0011.

Bradley, M. M., Miccoli, L., Escrig, M. A., & Lang, P. J. (2008). The pupil as a measure of emotional arousal and autonomic activation. *Psychophysiology, 45*(4), 602–607. https://doi.org/10.1111/j.1469-8986.2008.00654.x.

Burnette, J. L., McCullough, M. E., Van Tongeren, D. R., & Davis, D. E. (2012). Forgiveness results from integrating information about relationship value and exploitation risk. *Personality & Social Psychology Bulletin, 38*(3), 345–356. https://doi.org/10.1177/0146167211424582.

Callaghan, T., Moll, H., Rakoczy, H., Warneken, F., Liszkowski, U., Behne, T., ... Collins, W. A. (2011). Early social cognition in three cultural contexts. *Monographs of the Society for Research in Child Development, 76*(2), i–142.

Campos, J. J., Campos, R. G., & Barrett, K. C. (1989). Emergent themes in the study of emotional development and emotion regulation. *Developmental Psychology, 25*(3), 394–402. https://doi.org/10.1037/0012-1649.25.3.394.

Chai, Q., & He, J. (2017). Chinese preschoolers' resource allocation in the face of existing inequality under collaborative and noncollaborative contexts. *Developmental Psychology, 53*, 1494–1500. https://doi.org/10.1037/dev0000359.

Chai, Q., Yin, J., Shen, M., & He, J. (2024). Act generously when others do so: Majority influence on young children's sharing behavior. *Developmental Science, 27*, e13472. https://doi.org/10.1111/desc.13472.

Cheng, X., Wang, L., Fang, M., & Chen, Z. (2025). When do children forgive: The impact of the transgressor's intention and authority on children's forgiveness. *Early Childhood Education Journal*. https://doi.org/10.1007/s10643-025-02007-5.

Chiarella, S. S., & Poulin-Dubois, D. (2013). Cry babies and pollyannas: Infants can detect unjustified emotional reactions. *Infancy: The Official Journal of the International Society on Infant Studies, 18*(1), E81–E96. https://doi.org/10.1111/infa.12028.

Chiarella, S. S., & Poulin-Dubois, D. (2015). "Aren't you supposed to be sad?" Infants do not treat a stoic person as an unreliable emoter. *Infant Behavior & Development, 38*, 57–66. https://doi.org/10.1016/j.infbeh.2014.12.007.

Clayton, S., & Vaish, A. (2026). How sorry? Young children's forgiveness is sensitive to transgressors' depth of remorse. *Journal of Experimental Child Psychology, 266*, 106482. https://doi.org/10.1016/j.jecp.2026.106482.

Clore, G. L., & Huntsinger, J. R. (2007). How emotions inform judgment and regulate thought. *Trends in Cognitive Sciences, 11*(9), 393–399. https://doi.org/10.1016/j.tics.2007.08.005.

Dahl, A., Satlof-Bedrick, E. S., Hammond, S. I., Drummond, J. K., Waugh, W. E., & Brownell, C. A. (2017). Explicit scaffolding increases simple helping in younger infants. *Developmental Psychology, 53*(3), 407–416. https://doi.org/10.1037/dev0000244.

Darby, B. W., & Schlenker, B. R. (1989). Children's reactions to transgressions: Effects of the actor's apology, reputation and remorse. *The British Journal of Social Psychology, 28*(Pt 4), 353–364. https://doi.org/10.1111/j.2044-8309.1989.tb00879.x.

Davidov, M., Roth-Hanania, R., Paz, Y., Orlitsky, T., Uzefovsky, F., & Zahn-Waxler, C. (2025). Empathy development from birth to three: Advances in knowledge from 2000 to 2025. *Infant Behavior and Development, 81*, 102144. https://doi.org/10.1016/j.infbeh.2025.102144.

Davis, J. R., & Gold, G. J. (2011). An examination of emotional empathy, attributions of stability, and the link between perceived remorse and forgiveness. *Personality and Individual Differences, 50*(3), 392–397. https://doi.org/10.1016/j.paid.2010.10.031.

Deci, E. L., Koestner, R., & Ryan, R. M. (1999). A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. *Psychological Bulletin, 125*(6), 627–668. https://doi.org/10.1037/0033-2909.125.6.627.

Doosje, B., Branscombe, N. R., Spears, R., & Manstead, A. S. R. (1998). Guilty by association: When one's group has a negative history. *Journal of Personality and Social Psychology, 75*(4), 872–886. https://doi.org/10.1037/0022-3514.75.4.872.

Drell, M. B., & Jaswal, V. K. (2016). Making amends: Children's expectations about and responses to apologies. *Social Development, 25*(4), 742–758. https://doi.org/10.1111/sode.12168.

Drummond, J. D. K., Hammond, S. I., Satlof-Bedrick, E., Waugh, W. E., & Brownell, C. A. (2017). Helping the one you hurt: Toddlers' rudimentary guilt, shame, and prosocial behavior after harming another. *Child Development, 88*(4), 1382–1397. https://doi.org/10.1111/cdev.12653.

Dunfield, K. A., & Kuhlmeier, V. A. (2010). Intention-mediated selective helping in infancy. *Psychological Science, 21*(4), 523–527. https://doi.org/10.1177/0956797610364119.

Dunn, E. W., Aknin, L. B., & Norton, M. I. (2008). Spending money on others promotes happiness. *Science (New York, N. Y.), 319*(5870), 1687–1688. https://doi.org/10.1126/science.1150952.

Eisenberg, N., & Fabes, R. A. (1990). Empathy: Conceptualization, measurement, and relation to prosocial behavior. *Motivation and Emotion, 14*(2), 131–149. https://doi.org/10.1007/BF00991640.

Eisenberg, N., Spinrad, T., & Sadovsky, A. (2006). Empathy-related responding in children. *Handbook of moral development*. Psychology Press.

Eisenberg, N., VanSchyndel, S. K., & Spinrad, T. L. (2016). Prosocial motivation: Inferences from an opaque body of work. *Child Development, 87*(6), 1668–1678. https://doi.org/10.1111/cdev.12638.

Engelmann, J. M., & Rapp, D. J. (2018). The influence of reputational concerns on children's prosociality. *Current Opinion in Psychology, 20*, 92–95. https://doi.org/10.1016/j.copsyc.2017.08.024.

Enright, R. D., Gassin, E. A., & Wu, C. (1992). Forgiveness: A developmental view. *Journal of Moral Education, 21*(2), 99–114. https://doi.org/10.1080/0305724920210202.

Evans, A. D., & Lee, K. (2013). Emergence of lying in very young children. *Developmental Psychology, 49*(10), 1958–1963. https://doi.org/10.1037/a0031409.

Fehr, R., & Gelfand, M. J. (2010). When apologies work: How matching apology components to victims' self-construals facilitates forgiveness. *Organizational Behavior and Human Decision Processes, 113*(1), 37–50. https://doi.org/10.1016/j.obhdp.2010.04.002.

Fehr, R., Gelfand, M. J., & Nag, M. (2010). The road to forgiveness: A meta-analytic synthesis of its situational and dispositional correlates. *Psychological Bulletin, 136*(5), 894–914. https://doi.org/10.1037/a0019993.

Gerdemann, S. C., Vaish, A., & Hepach, R. (2025). Body posture as a measure of emotional valence in young children: A preregistered validation study. *Frontiers in Developmental Psychology, 3*. https://doi.org/10.3389/fdpys.2025.1536440.

Gleason, J. B., & Weintraub, S. (1976). The acquisition of routines in child language. *Language in Society, 5*(2), 129–136. https://doi.org/10.1017/S0047404500006977.

Graham, S. (1988). Children's developing understanding of the motivational role of affect: An attributional analysis. *Cognitive Development, 3*(1), 71–88. https://doi.org/10.1016/0885-2014(88)90031-7.

Grossmann, T., Missana, M., & Krol, K. M. (2018). The neurodevelopmental precursors of altruistic behavior in infancy. *PLoS Biology, 16*(9), e2005281. https://doi.org/10.1371/journal.pbio.2005281.

Grueneisen, S., & Warneken, F. (2022). The development of prosocial behavior-from sympathy to strategy. *Current Opinion in Psychology, 43*, 323–328. https://doi.org/10.1016/j.copsyc.2021.08.005.

Hamann, K., Warneken, F., Greenberg, J., & Tomasello, M. (2011). Collaboration encourages equal sharing in children but not in chimpanzees. *Nature, 476*, 328–331. https://doi.org/10.1038/nature10278.

Harbaugh, W. T., Mayr, U., & Burghart, D. R. (2007). Neural responses to taxation and voluntary giving reveal motives for charitable donations. *Science (New York, N. Y.), 316*(5831), 1622–1625. https://doi.org/10.1126/science.1140738.

Hay, D. F. (1979). Cooperative interactions and sharing between very young children and their parents. *Developmental Psychology, 15*(6), 647–653. https://doi.org/10.1037/0012-1649.15.6.647.

Henrich, J. (2020). *The WEIRDest people in the world: How the West became psychologically peculiar and particularly prosperous*. Farrar, Straus and Giroux.

Henrich, J., Heine, S. J., & Norenzayan, A. (2010). The weirdest people in the world? *The Behavioral and Brain Sciences, 33*(2–3), 61–83. https://doi.org/10.1017/S0140525×0999152X discussion 83-135.

Hepach, R., Engelmann, J. M., Herrmann, E., Gerdemann, S. C., & Tomasello, M. (2023). Evidence for a developmental shift in the motivation underlying helping in early childhood. *Developmental Science, 26*(1), e13253. https://doi.org/10.1111/desc.13253.

Hepach, R., & Vaish, A. (2020). The study of prosocial emotions in early childhood: Unique opportunities and insights. *Emotion Review, 12*(4), 278–279. https://doi.org/10.1177/1754073920939630.

Hepach, R., Vaish, A., Grossmann, T., & Tomasello, M. (2016). Young children want to see others get the help they need. *Child Development, 87*(6), 1703–1714. https://doi.org/10.1111/cdev.12633.

Hepach, R., Vaish, A., Müller, K., & Tomasello, M. (2019). The relation between young children's physiological arousal and their motivation to help others. *Neuropsychologia, The Biological Basis of Social Cognition During Development, 126*, 113–119. https://doi.org/10.1016/j.neuropsychologia.2017.10.010.

Hepach, R., Vaish, A., & Tomasello, M. (2012). Young children are intrinsically motivated to see others helped. *Psychological Science, 23*, 967–972. https://doi.org/10.1177/0956797612440571.

Hepach, R., Vaish, A., & Tomasello, M. (2017a). Children's intrinsic motivation to provide help themselves after accidentally harming others. *Child Development, 88*(4), 1251–1264. https://doi.org/10.1111/cdev.12646.

Hepach, R., Vaish, A., & Tomasello, M. (2017b). The fulfillment of others' needs elevates children's body posture. *Developmental Psychology, 53*(1), 100–113. https://doi.org/10.1037/dev0000173.

Hepach, R., & Westermann, G. (2016). Pupillometry in infancy research. *Journal of Cognition and Development, 17*. https://doi.org/10.1080/15248372.2015.1135801.

Hoffman, M. L. (1982). Affect and moral development. *New Directions for Child and Adolescent Development, 1982*(16), 83–103. https://doi.org/10.1002/cd.23219821605.

Hoffman, M. L. (2000). *Empathy and moral development: Implications for caring and justice*. Cambridge University Press. 331. https://doi.org/10.1017/CBO9780511805851 x.

Hoffman, M. L. (2007). The origins of empathic morality in toddlerhood. *Socioemotional development in the toddler years: Transitions and transformations*. The Guilford Press. 132–145.

House, B. R. (2018). How do social norms influence prosocial development? *Current Opinion in Psychology, 20*, 87–91. https://doi.org/10.1016/j.copsyc.2017.08.011.

Huang, H., Su, Y., & Jin, J. (2017). Empathy-related responding in Chinese toddlers: Factorial structure and cognitive contributors. *Infant and Child Development, 26*(3), e1983. https://doi.org/10.1002/icd.1983.

Johanson, D., & Edey, M. (1990). *Lucy: The beginnings of humankind*. Simon and Schuster.

Jordan, M. R., Amir, D., & Bloom, P. (2016). Are empathy and concern psychologically distinct? *Emotion (Washington, D. C.), 16*(8), 1107–1116. https://doi.org/10.1037/emo0000228.

Kärtner, J., Crafa, D., Chaudhary, N., & Keller, H. (2016). Reactions to receiving a gift-maternal scaffolding and cultural learning in Berlin and Delhi. *Child Development, 87*(3), 712–722. https://doi.org/10.1111/cdev.12525.

Kärtner, J., Keller, H., & Chaudhary, N. (2010). Cognitive and social influences on early prosocial behavior in two sociocultural contexts. *Developmental Psychology, 46*(4), 905–914. https://doi.org/10.1037/a0019718.

Kelsey, C., Grossmann, T., & Vaish, A. (2018). Early reputation management: Three-year-old children are more generous following exposure to eyes. *Frontiers in Psychology, 9*. https://doi.org/10.3389/fpsyg.2018.00698.

Keltner, D. (1995). Signs of appeasement: Evidence for the distinct displays of embarrassment, amusement, and shame. *Journal of Personality and Social Psychology, 68*(3), 441–454. https://doi.org/10.1037/0022-3514.68.3.441.

Keltner, D., Haidt, J., & Shiota, M. N. (2006). Social functionalism and the evolution of emotions. *Evolution and social psychology*. Psychology Press.

Keltner, D., Horberg, E. J., & Oveis, C. (2006). Emotions as moral intuitions. *Affect in social thinking and behavior*. Psychology Press.

Keltner, D., Young, R. C., & Buswell, B. N. (1997). Appeasement in human emotion, social practice, and personality. *Aggressive Behavior, 23*(5), 359–374. https://doi.org/10.1002/(SICI)1098-2337(1997)23:5%3C359::AID-AB5%3E3.0.CO;2-D.

Ketelaar, T., & Tung Au, W. (2003). The effects of feelings of guilt on the behaviour of uncooperative individuals in repeated social bargaining games: An affect-as-information interpretation of the role of emotion in social interaction. *Cognition & Emotion, 17*(3), 429–453. https://doi.org/10.1080/02699930143000662.

Kochanska, G., Casey, R. J., & Fukumoto, A. (1995). Toddlers' sensitivity to standard violations. *Child Development, 66*(3), 643–656. https://doi.org/10.2307/1131940.

Leach, C. W., & Cidam, A. (2015). When is shame linked to constructive approach orientation? A meta-analysis. *Journal of Personality and Social Psychology, 109*(6), 983–1002. https://doi.org/10.1037/pspa0000037.

Ma, L. K., Tunney, R. J., & Ferguson, E. (2017). Does gratitude enhance prosociality?: A meta-analytic review. *Psychological Bulletin, 143*(6), 601–635. https://doi.org/10.1037/bul0000103.

Martin, A., Lin, K., & Olson, K. R. (2016). What you want versus what's good for you: Paternalistic motivation in children's helping behavior. *Child Development, 87*(6), 1739–1746. https://doi.org/10.1111/cdev.12637.

Martin, A., & Olson, K. R. (2013). When kids know better: Paternalistic helping in 3-year-old children. *Developmental Psychology, 49*(11), 2071–2081. https://doi.org/10.1037/a0031715.

McAuliffe, K., Raihani, N. J., & Dunham, Y. (2017). Children are sensitive to norms of giving. *Cognition, 167*, 151–159. https://doi.org/10.1016/j.cognition.2017.01.006.

McCullough, M. E., Kilpatrick, S. D., Emmons, R. A., & Larson, D. B. (2001). Is gratitude a moral affect? *Psychological Bulletin, 127*(2), 249–266. https://doi.org/10.1037/0033-2909.127.2.249.

McCullough, M. E., Kimeldorf, M. B., & Cohen, A. D. (2008). An adaptation for altruism? The social causes, social effects, and social evolution of gratitude. *Current Directions in Psychological Science, 17*(4), 281–285. https://doi.org/10.1111/j.1467-8721.2008.00590.x.

McCullough, M. E., Kurzban, R., & Tabak, B. A. (2013). Cognitive systems for revenge and forgiveness. *Behavioral and Brain Sciences, 36*(1), 1–15. https://doi.org/10.1017/S0140525×11002160.

McCullough, M. E., Pedersen, E. J., Tabak, B. A., & Carter, E. C. (2014). Conciliatory gestures promote forgiveness and reduce anger in humans. *Proceedings of the National Academy of Sciences, 111*(30), 11211–11216. https://doi.org/10.1073/pnas.1405072111.

McCullough, M. E., Worthington, E. L., & Rachal, K. C. (1997). Interpersonal forgiving in close relationships. *Journal of Personality and Social Psychology, 73*(2), 321–336. https://doi.org/10.1037//0022-3514.73.2.321.

McElroy, C. E., Kelsey, C. M., Oostenbroek, J., & Vaish, A. (2023). Beyond accidents: Young children's forgiveness of third-party intentional transgressors. *Journal of Experimental Child Psychology, 228*, 105607. https://doi.org/10.1016/j.jecp.2022.105607.

McLaughlin, A., Marshall, J., Gonzalez-Rubio Saab, I., & McAuliffe, K. (2025). Children's evaluations and expectations of forgiveness following second- and third-party interventions. *Child Development, 96*(6), 2146–2161. https://doi.org/10.1111/cdev.70030.

McLaughlin, A., Marshall, J., & McAuliffe, K. (2024). Developing conceptions of forgiveness across the lifespan. *Child Development, 95*(6), 1915–1933. https://doi.org/10.1111/cdev.14122.

Miller, J. G. (2018). Physiological mechanisms of prosociality. *Current Opinion in Psychology, 20*, 50–54. https://doi.org/10.1016/j.copsyc.2017.08.018.

Molho, C., Soraperra, I., Schulz, J. F., & Shalvi, S. (2025). Guilt drives prosociality across 20 countries. *Nature Human Behaviour, 9*(10), 2199–2211. https://doi.org/10.1038/s41562-025-02286-3.

Mulvey, K. L., Gönültaş, S., Herry, E., & Strelan, P. (2022). The role of theory of mind, group membership, and apology in intergroup forgiveness among children and adolescents. *Journal of Experimental Psychology: General, 151*(3), 613–627. https://doi.org/10.1037/xge0001094.

Nelissen, R. M. A., & Zeelenberg, M. (2009). When guilt evokes self-punishment: Evidence for the existence of a Dobby effect. *Emotion (Washington, D. C.), 9*(1), 118–122. https://doi.org/10.1037/a0014540.

Nesse, R. M. (1990). Evolutionary explanations of emotions. *Human Nature, 1*(3), 261–289. https://doi.org/10.1007/BF02733986.

Newton, E. K., Thompson, R. A., & Goodman, M. (2016). Individual differences in toddlers' prosociality: Experiences in early relationships explain variability in prosocial behavior. *Child Development, 87*(6), 1715–1726. https://doi.org/10.1111/cdev.12631.

Nowak, M. A., & Roch, S. (2006). Upstream reciprocity and the evolution of gratitude. *Proceedings of the Royal Society B: Biological Sciences, 274*(1610), 605–610. https://doi.org/10.1098/rspb.2006.0125.

Nowak, M. A., & Sigmund, K. (2005). Evolution of indirect reciprocity. *Nature, 437*(7063), 1291–1298. https://doi.org/10.1038/nature04131.

Ohtsubo, Y., & Watanabe, E. (2009). Do sincere apologies need to be costly? Test of a costly signaling model of apology. *Evolution and Human Behavior, 30*(2), 114–123. https://doi.org/10.1016/j.evolhumbehav.2008.09.004.

Oostenbroek, J., & Vaish, A. (2019a). The benefits of forgiving: Young children respond positively to those who forgive. *Journal of Experimental Psychology. General, 148*(11), 1914–1924. https://doi.org/10.1037/xge0000576.

Oostenbroek, J., & Vaish, A. (2019b). The emergence of forgiveness in young children. *Child Development, 90*(6), 1969–1986. https://doi.org/10.1111/cdev.13069.

Over, H., Vaish, A., & Tomasello, M. (2016). Do young children accept responsibility for the negative actions of ingroup members? *Cognitive Development, 40*, 24–32. https://doi.org/10.1016/j.cogdev.2016.08.004.

Paulus, M., & Moore, C. (2017). Preschoolers' generosity increases with understanding of the affective benefits of sharing. *Developmental Science, 20*(3), e12417. https://doi.org/10.1111/desc.12417.

Poelker, K. E., & Kuebli, J. E. (2014). Does the thought count? Gratitude understanding in elementary school students. *The Journal of Genetic Psychology, 175*(5–6), 431–448. https://doi.org/10.1080/00221325.2014.941321.

Pons, F., Lawson, J., Harris, P. L., & de Rosnay, M. (2003). Individual differences in children's emotion understanding: Effects of age and language. *Scandinavian Journal of Psychology, 44*(4), 347–353. https://doi.org/10.1111/1467-9450.00354.

Rheingold, H. L., Hay, D. F., & West, M. J. (1976). Sharing in the second year of life. *Child Development, 47*(4), 1148–1158. https://doi.org/10.2307/1128454.

Rizzo, M. T., & Killen, M. (2016). Children's understanding of equity in the context of inequality. *British Journal of Developmental Psychology, 34*(4), 569–581. https://doi.org/10.1111/bjdp.12150.

Ross, J. (2017). You and me: Investigating the role of self-evaluative emotion in preschool prosociality. *Journal of Experimental Child Psychology, 155*, 67–83. https://doi.org/10.1016/j.jecp.2016.11.001.

Rottman, J. (2014). Evolution, development, and the emergence of disgust. *Evolutionary Psychology, 12*(2), 417–433. https://doi.org/10.1177/147470491401200209.

Rozin, P., & Fallon, A. E. (1987). A perspective on disgust. *Psychological Review, 94*(1), 23–41. https://doi.org/10.1037/0033-295X.94.1.23.

Scher, S. J., & Darley, J. M. (1997). How effective are the things people say to apologize? Effects of the realization of the apology speech act. *Journal of Psycholinguistic Research, 26*(1), 127–140. https://doi.org/10.1023/A:1025068306386.

Schmidt, M. F. H., Rakoczy, H., & Tomasello, M. (2019). Eighteen-month-old infants correct non-conforming actions by others. *Infancy: The Official Journal of the International Society on Infant Studies, 24*(4), 613–635. https://doi.org/10.1111/infa.12292.

Scott-Phillips, T. C., Dickins, T. E., & West, S. A. (2011). Evolutionary theory and the ultimate–proximate distinction in the human behavioral sciences. *Perspectives on Psychological Science, 6*(1), 38–47. https://doi.org/10.1177/1745691610393528.

Shaw, A., & Olson, K. R. (2012). Children discard a resource to avoid inequity. *Journal of Experimental Psychology: General, 141*(2), 382–395. https://doi.org/10.1037/a0025907.

Shoshani, A., Aharon-Dvir, O., Hain, D., & Yaffe, A. (2021). Situational determinants of young children's gratitude: The effects of perceived intentionality and the value of the benefit on gratitude and prosocial behavior. *Journal of Personality and Social Psychology, 121*(4), 914–932. https://doi.org/10.1037/pspp0000384.

Shoshani, A., De-Leon Lendner, K., Nissensohn, A., Lazarovich, G., & Aharon-Dvir, O. (2020). Grateful and kind: The prosocial function of gratitude in young children's relationships. *Developmental Psychology, 56*(6), 1135–1148. https://doi.org/10.1037/dev0000922.

Sivertsen, S. S., Haun, D., & Hepach, R. (2024). Choosing to help others at a cost to oneself elevates preschoolers' body posture. *Evolution and Human Behavior, 45*(2), 175–182. https://doi.org/10.1016/j.evolhumbehav.2024.02.001.

Small, D. A., & Loewenstein, G. (2003). Helping a victim or helping the victim: Altruism and identifiability. *Journal of Risk and Uncertainty, 26*(1), 5–16. https://doi.org/10.1023/A:1022299422219.

Smith, C. E., Blake, P. R., & Harris, P. L. (2013). I should but i won't: Why young children endorse norms of fair sharing but do not follow them. *PLoS One, 8*(3), e59510. https://doi.org/10.1371/journal.pone.0059510.

Smith, C. E., Chen, D., & Harris, P. L. (2010). When the happy victimizer says sorry: Children's understanding of apology and emotion. *British Journal of Developmental Psychology, 28*(4), 727–746. https://doi.org/10.1348/026151009×475343.

Smith, C. E., & Harris, P. L. (2012). He didn't want me to feel sad: Children's reactions to disappointment and apology. *Social Development, 21*(2), 215–228. https://doi.org/10.1111/j.1467-9507.2011.00606.x.

Snippe, E., Jeronimus, B. F., Aan Het Rot, M., Bos, E. H., de Jonge, P., & Wichers, M. (2018). The reciprocity of prosocial behavior and positive affect in daily life. *Journal of Personality, 86*(2), 139–146. https://doi.org/10.1111/jopy.12299.

Song, Y., Broekhuizen, M. L., & Dubas, J. S. (2020). Happy little benefactor: Prosocial behaviors promote happiness in young children from two cultures. *Frontiers in Psychology, 11*. https://doi.org/10.3389/fpsyg.2020.01398.

Struthers, C. W., Eaton, J., Santelli, A. G., Uchiyama, M., & Shirvani, N. (2008). The effects of attributions of intent and apology on forgiveness: When saying sorry may not help the story. *Journal of Experimental Social Psychology, 44*(4), 983–992. https://doi.org/10.1016/j.jesp.2008.02.006.

Svetlova, M., Nichols, S. R., & Brownell, C. A. (2010). Toddlers' prosocial behavior: From instrumental to empathic to altruistic helping. *Child Development, 81*(6), 1814–1827. https://doi.org/10.1111/j.1467-8624.2010.01512.x.

Tang, M., Li, W., LIU, F., & Yuan, B. (2019). The association between guilt and prosocial behavior: A systematic review and meta-analysis. *Advances in Psychological Science, 27*, 773. https://doi.org/10.3724/SP.J.1042.2019.00773.

Tangney, J. P., & Dearing, R. L. (2002). *Shame and guilt*. Guilford Press. xvi, 272. https://doi.org/10.4135/9781412950664.n388.

Toda, R., Toda, N., Hagihara, H., & Kanakogi, Y. (2024). Understanding of the functions of forgiveness among preschoolers. *Journal of Experimental Child Psychology, 247*, 106036. https://doi.org/10.1016/j.jecp.2024.106036.

Tomasello, M. (2016). *A natural history of human morality*. Harvard University Press. x, 194.

Tomasello, M. (2018). Precís of a natural history of human morality. *Philosophical Psychology, 31*(5), 661–668. https://doi.org/10.1080/09515089.2018.1486605.

Tooby, J., & Cosmides, L. (1990). The past explains the present: Emotional adaptations and the structure of ancestral environments. *Ethology and Sociobiology, 11*(4), 375–424. https://doi.org/10.1016/0162-3095(90)90017-Z.

Tracy, J. L., Robins, R. W., & Lagattuta, K. H. (2005). Can children recognize pride? *Emotion (Washington, D. C.), 5*(3), 251–257. https://doi.org/10.1037/1528-3542.5.3.251.

Trivers, R. L. (1971). The evolution of reciprocal altruism. *The Quarterly Review of Biology, 46*(1), 35–57. https://doi.org/10.1086/406755.

Trommsdorff, G., Friedlmeier, W., & Mayer, B. (2007). Sympathy, distress, and prosocial behavior of preschool children in four cultures. *International Journal of Behavioral Development, 31*(3), 284–293. https://doi.org/10.1177/0165025407076441.

Tsai, J. L., Knutson, B., & Fung, H. H. (2006). Cultural variation in affect valuation. *Journal of Personality and Social Psychology, 90*(2), 288–307. https://doi.org/10.1037/0022-3514.90.2.288.

Tsang, J.-A. (2006a). BRIEF REPORT gratitude and prosocial behaviour: An experimental test of gratitude. *Cognition and Emotion, 20*(1), 138–148. https://doi.org/10.1080/02699930500172341.

Tsang, J.-A. (2006b). The effects of helper intention on gratitude and indebtedness. *Motivation and Emotion, 30*(3), 199–205. https://doi.org/10.1007/s11031-006-9031-z.

Tsang, J.-A. (2007). Gratitude for small and large favors: A behavioral test. *The Journal of Positive Psychology, 2*(3), 157–167. https://doi.org/10.1080/17439760701229019.

Tybur, J. M., Lieberman, D., Kurzban, R., & DeScioli, P. (2013). Disgust: Evolved function and structure. *Psychological Review, 120*(1), 65–84. https://doi.org/10.1037/a0030778.

Ulber, J., Hamann, K., & Tomasello, M. (2016). Extrinsic rewards diminish costly sharing in 3-year-olds. *Child Development, 87*(4), 1192–1203. https://doi.org/10.1111/cdev.12534.

Vaish, A. (2016). Flexible concern: The development of multidetermined and context-dependent empathic responding. *Child Development Perspectives, 10*(3), 149–154. https://doi.org/10.1111/cdep.12178.

Vaish, A. (2018). The prosocial functions of early social emotions: The case of guilt. *Current Opinion in Psychology, 20*, 25–29. https://doi.org/10.1016/j.copsyc.2017.08.008.

Vaish, A., Carpenter, M., & Tomasello, M. (2009). Sympathy through affective perspective taking and its relation to prosocial behavior in toddlers. *Developmental Psychology, 45*(2), 534–543. https://doi.org/10.1037/a0014322.

Vaish, A., Carpenter, M., & Tomasello, M. (2011). Young children's responses to guilt displays. *Developmental Psychology, 47*(5), 1248–1262. https://doi.org/10.1037/a0024462.

Vaish, A., Carpenter, M., & Tomasello, M. (2016). The early emergence of guilt-motivated prosocial behavior. *Child Development, 87*(6), 1772–1782. https://doi.org/10.1111/cdev.12628.

Vaish, A., & Hepach, R. (2020). The development of prosocial emotions. *Emotion Review, 12*(4), 259–273. https://doi.org/10.1177/1754073919885014.

Vaish, A., Hepach, R., & Tomasello, M. (2018). The specificity of reciprocity: Young children reciprocate more generously to those who intentionally benefit them. *Journal of Experimental Child Psychology, 167*, 336–353. https://doi.org/10.1016/j.jecp.2017.11.005.

Vaish, A., & Oostenbroek, J. (2022). Preferential forgiveness: The impact of group membership and remorse on preschoolers' forgiveness. *Journal of Experimental Psychology. General, 151*(5), 1132–1140. https://doi.org/10.1037/xge0001114.

Vaish, A., & Savell, S. (2022). Young children value recipients who display gratitude. *Developmental Psychology, 58*(4), 680–692. https://doi.org/10.1037/dev0001308.

Vaish, A., & Tomasello, M. (2022). The early ontogeny of human cooperation and morality. *Handbook of moral development* (3rd ed.). Routledge.

van Kleef, G. A., & Lelieveld, G.-J. (2022). Moving the self and others to do good: The emotional underpinnings of prosocial behavior. *Current Opinion in Psychology, 44*, 80–88. https://doi.org/10.1016/j.copsyc.2021.08.029.

Vreden, C., Buryn-Weitzel, J. C., Atim, S., Donnellan, E., Hoffman, M., Holden, E., ... Clay, Z. (2025). Early empathy development: Concern and comforting in 9- and 18-month-old infants from Uganda and the UK. *PLoS One, 20*(5), e0320371. https://doi.org/10.1371/journal.pone.0320371.

Waddington, O., Jensen, K., & Köymen, B. (2022). Boundaries of apologies: Children avoid transgressors who give the same apology for a repeat offence. *Cognitive Development, 64*, 101264. https://doi.org/10.1016/j.cogdev.2022.101264.

Waddington, O., Proft, M., Jensen, K., & Köymen, B. (2023). Five-year-old children value reasons in apologies for belief-based accidents. *Child Development, 94*(3), e143–e153. https://doi.org/10.1111/cdev.13893.

Warneken, F., Hare, B., Melis, A. P., Hanus, D., & Tomasello, M. (2007). Spontaneous altruism by chimpanzees and young children. *PLoS Biology, 5*(7), e184. https://doi.org/10.1371/journal.pbio.0050184.

Warneken, F., & Tomasello, M. (2006). Altruistic helping in human infants and young chimpanzees. *Science (New York, N. Y.), 311*(5765), 1301–1303. https://doi.org/10.1126/science.1121448.

Warneken, F., & Tomasello, M. (2008). Extrinsic rewards undermine altruistic tendencies in 20-month-olds. *Developmental Psychology, 44*(6), 1785–1788. https://doi.org/10.1037/a0013860.

Warneken, F., & Tomasello, M. (2013). The emergence of contingent reciprocity in young children. *Journal of Experimental Child Psychology, 116*(2), 338–350. https://doi.org/10.1016/j.jecp.2013.06.002.

Young, G., & Lewis, M. (1979). Effects of familiarity and maternal attention on infant peer relations. *Merrill-Palmer Quarterly, 25*(2), 105–119.

Zahn-Waxler, C., Radkerrow, M., Wagner, E., & Chapman, M. (1992). Development of concern for others. *Developmental Psychology, 28*, 126–135.

Zaki, J., Schirmer, J., & Mitchell, J. P. (2011). Social influence modulates the neural computation of value. *Psychological Science, 22*(7), 894–900. https://doi.org/10.1177/0956797611411057.

Zhang, X., & Li, Y. (2026). The ontogeny of children's group-based guilt and motivated reparative prosocial behaviors. *Emotion (Washington, D. C.), 26*(1), 113–127. https://doi.org/10.1037/emo0001557.

Zhang, Z., & Benozio, A. (2021). Waste aversion reduces inequity aversion among chinese children. *Child Development, 92*(6), 2465–2477. https://doi.org/10.1111/cdev.13586.